

Physical Chemistry (I)

Lecture 23

Statistical Mechanics

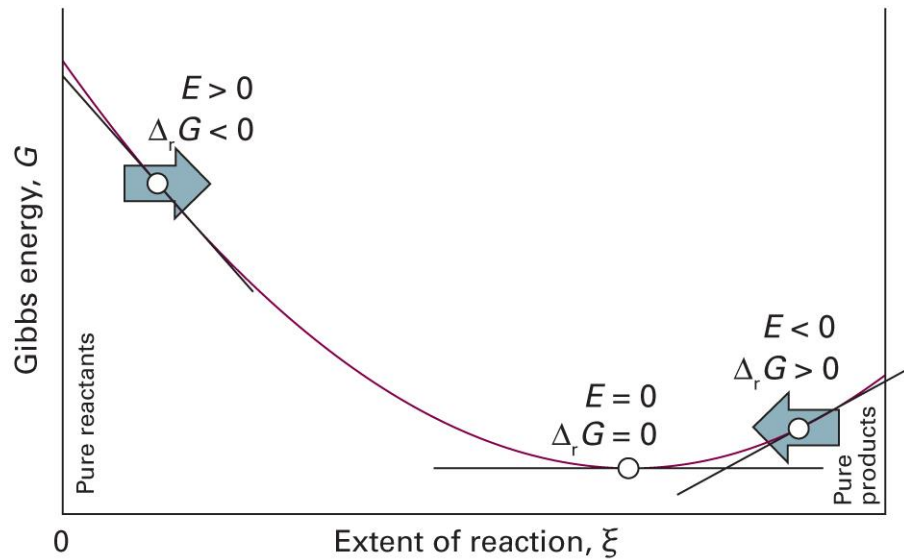


In the Last Class

- Electrochemistry
 - Thermodynamics of cell
 - Thermodynamics of electrodes
- Statistical mechanics
 - Boltzmann factor and partition function
 - Classical statistical mechanics: phase space

Thermodynamics of Cell

$$\Delta_r G = -\nu F E_{\text{cell}}$$



Last class

Nernst Equation

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

$$E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln K$$

Electrode Potentials

Cell potential = cathode contribution + anode contribution

For a cell L||R,

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{R}) - E^{\ominus}(\text{L})$$

- Standard potential $E^{\ominus}(\text{X})$

For a cell $\text{Pt}(\text{s}) | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{X}$, $E^{\ominus}(\text{X}) = E_{\text{cell}}^{\ominus}$

Last class

Thermodynamics of Potentials

$$\Delta_r G^\ominus(c) = \Delta_r G^\ominus(a) + \Delta_r G^\ominus(b)$$

$$\nu_c E^\ominus(c) = \nu_a E^\ominus(a) + \nu_b E^\ominus(b)$$

Boltzmann Factor

In general, for a microstate s with energy E ,

$$P(s) \propto e^{-E(s)/k_B T} \text{ (Boltzmann factor)}$$

or, using a constant q ,

$$P(s) = \frac{1}{q} e^{-E(s)/k_B T}$$

$$q = \sum_s e^{-E(s)/k_B T}$$

(Partition function)

Phase Space

Space of **all possible microstates** of a system

- Classical mechanics:

Space of all possible values of
position and **momentum** variables

Partition Function

$$q = \frac{1}{h} \int \int e^{-E(x,p)/k_B T} dx dp$$

- Example: 1-dimensional spring

$$E(x, p) = \frac{1}{2} k x^2 + \frac{1}{2} m v^2 = \frac{1}{2} k x^2 + \frac{p^2}{2m}$$

$$q = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{k_B T} \left(\frac{1}{2} k x^2 + \frac{p^2}{2m} \right)} dx dp = \frac{2\pi k_B T}{h} \sqrt{\frac{m}{k}}$$

Into Higher Dimensions

- Single 1-dimensional particle

$$q = \frac{1}{h} \int \int e^{-E(x,p)/k_B T} dx dp$$

- Single 3-dimensional particle

$$q = \frac{1}{h^3} \iiint \iiint e^{-E(x,y,z,p_x,p_y,p_z)/k_B T} dx dy dz dp_x dp_y dp_z$$

$$\rightarrow q = \frac{1}{h^3} \int \int e^{-E(\vec{r},\vec{p})/k_B T} d\vec{r} d\vec{p}$$

Higher and Higher

- Single 3-dimensional particle

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

- N 3-dimensional (independent) particles

$$Q = q_1 \times q_2 \times q_3 \times q_4 \times \cdots \times q_N$$

$$Q = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N$$

Indistinguishability

- N 3-dimensional distinguishable particles:

$$Q = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N$$

- N 3-dimensional indistinguishable particles:

$$Q = \frac{1}{N! h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N$$

Probability Distributions

	Discrete	Continuous
Variable	$k = 1, 2, \dots, N$	x : real in $[x_1, x_2]$
Probability	$p(k)$	$p(x)dx$ probability density ↘
Normalization	$\sum_{k=1}^N p(k) = 1$	$\int_{x_1}^{x_2} p(x) dx = 1$
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$	$\langle x \rangle = \int_{x_1}^{x_2} xp(x) dx$
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$	$\langle f \rangle = \int_{x_1}^{x_2} f(x)p(x) dx$

Classical Statistical Mechanics

For a single particle,

- Variable: $\vec{r}$ and $\vec{p}$
- Probability:

$$\frac{1}{h^3} P(\vec{r}, \vec{p}) d\vec{r} d\vec{p} = \frac{1}{h^3} \frac{e^{-E(\vec{r}, \vec{p})/k_B T}}{q} d\vec{r} d\vec{p}$$

- Normalization:

$$\int \int \frac{1}{h^3 q} e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} = \frac{1}{q} \times q = 1$$

Classical Statistical Mechanics

For a single particle,

- Probability:

$$\frac{1}{h^3} P(\vec{r}, \vec{p}) d\vec{r} d\vec{p} = \frac{1}{h^3 q} e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

- Expectation value:

$$\begin{aligned} \langle f \rangle &= \frac{1}{h^3} \int \int f(\vec{r}, \vec{p}) P(\vec{r}, \vec{p}) d\vec{r} d\vec{p} \\ &= \frac{1}{h^3 q} \int \int f(\vec{r}, \vec{p}) e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \end{aligned}$$

Internal Energy

- Internal energy is the expectation value of energy

$$U = \langle E \rangle = \frac{1}{h^3 q} \int \int E(\vec{r}, \vec{p}) e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Define a new variable: $\beta = 1/k_B T$

$$U = \frac{1}{h^3 q} \int \int E(\vec{r}, \vec{p}) e^{-\beta E(\vec{r}, \vec{p})} d\vec{r} d\vec{p}$$

Small Trick

$$U = \frac{1}{h^3 q} \int \int E(\vec{r}, \vec{p}) e^{-\beta E(\vec{r}, \vec{p})} d\vec{r} d\vec{p}$$

Since $\partial e^{-ax} / \partial x = -a e^{-ax}$,

$$U = -\frac{1}{h^3 q} \int \int \left(\frac{\partial e^{-\beta E(\vec{r}, \vec{p})}}{\partial \beta} \right) d\vec{r} d\vec{p}$$

$$U = -\frac{1}{q} \frac{\partial}{\partial \beta} \left(\frac{1}{h^3} \int \int e^{-\beta E(\vec{r}, \vec{p})} d\vec{r} d\vec{p} \right) = -\frac{1}{q} \frac{\partial q}{\partial \beta}$$

One More Trick

$$U = -\frac{1}{q} \frac{\partial q}{\partial \beta}$$

Using $\frac{\partial(\ln y)}{\partial x} = \frac{1}{y} \frac{\partial y}{\partial x},$

$$U = -\frac{\partial(\ln q)}{\partial \beta}$$

In general,

$$U = -\frac{\partial(\ln Q)}{\partial \beta}$$

Entropy

Consider a system with the following properties:

- t microstates
- each microstate i with probability p_i
- N independent replicates ($N \rightarrow \infty$)

Among N , the number of finding microstate i is $n_i = Np_i$

Then, the number of combinations is

$$W = \frac{N!}{n_1! n_2! \cdots n_t!}$$

Shannon Entropy

$$W = \frac{N!}{n_1! n_2! \cdots n_t!}$$

$$S = k_B \ln W = k_B \ln \frac{N!}{n_1! n_2! \cdots n_t!}$$

By Stirling's approximation,

$$S \approx k_B (N \ln N - \sum_i n_i \ln n_i) = -k_B \sum_i n_i \ln(n_i/N)$$

$$S = -Nk_B \sum_i p_i \ln p_i$$

Hence, entropy per system is $S = -k_B \sum_i p_i \ln p_i$

At Thermal Equilibrium

$$S = -k_B \sum_i p_i \ln p_i$$

Substitution of $p_i = \frac{1}{Q} e^{-\beta E_i}$ leads to

$$S = -k_B \sum_i \frac{e^{-\beta E_i}}{Q} (-\beta E_i - \ln Q)$$

$$S = \beta k_B \sum_i \frac{E_i e^{-\beta E_i}}{Q} + \frac{k_B \ln Q}{Q} \sum_i e^{-\beta E_i}$$

$$S = \frac{\langle E \rangle}{T} + k_B \ln Q$$

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Helmholtz Energy (?!)

$$S = \frac{\langle E \rangle}{T} + k_B \ln Q$$

Therefore,

$$-k_B T \ln Q = \langle E \rangle - TS = A$$

Partition Function and ...

- Helmholtz energy: $A = -k_{\text{B}}T \ln Q$
- Internal energy: $U = -\partial(\ln Q)/\partial\beta$
- Entropy: $S = (U - A)/T$

“A partition function contains
all the **thermodynamic information** about a system.”

Derived Functions

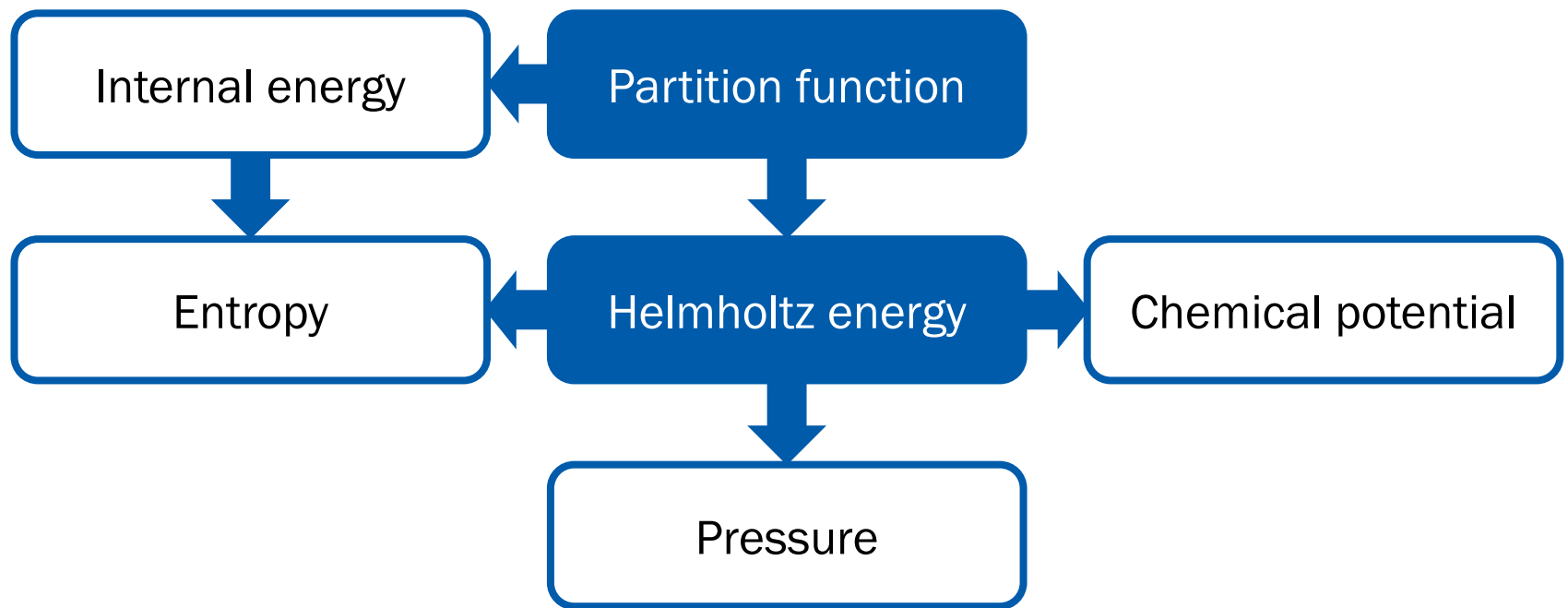
We know Helmholtz energy from the partition function,

$$A = -k_{\text{B}}T \ln Q$$

Using $dA = -SdT - pdV + \mu dn$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,n}, \quad p = -\left(\frac{\partial A}{\partial V}\right)_{T,n}, \quad \mu = \left(\frac{\partial A}{\partial n}\right)_{T,V}$$

Q and Thermodynamics



Calculation of Q

For indistinguishable particles,

$$Q = \frac{1}{N! h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N = \frac{1}{N!} q^N$$

where

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy ($p^2/2m$)
- Bound in a box of volume V
 - Range of $x = (0, a)$
 - Range of $y = (0, b)$
 - Range of $z = (0, c)$
 - $abc = V$
- Range of momentum = $(-\infty, \infty)$

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q of Perfect Gas

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Since $E(\vec{r}, \vec{p}) = p^2/2m$,

$$q = \frac{1}{h^3} \int \int e^{-p^2/2mk_B T} d\vec{r} d\vec{p}$$

$$q = \frac{1}{h^3} \left[\int 1 d\vec{r} \right] \left[\int e^{-p^2/2mk_B T} d\vec{p} \right]$$

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Position Integral

$$q = \frac{1}{h^3} \left[\int 1 \, d\vec{r} \right] \left[\int e^{-p^2/2mk_{\text{B}}T} \, d\vec{p} \right]$$

$$\int 1 \, d\vec{r} = \int_0^c \int_0^b \int_0^a 1 \, dx \, dy \, dz = abc = V$$

Momentum Integral

$$q = \frac{V}{h^3} \left[\int e^{-p^2/2mk_B T} d\vec{p} \right]$$

$$\int e^{-p^2/2mk_B T} d\vec{p} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-p^2/2mk_B T} dp_x dp_y dp_z$$

Since $e^{-p^2/2mk_B T} = e^{-p_x^2/2mk_B T} e^{-p_y^2/2mk_B T} e^{-p_z^2/2mk_B T}$,

$$\left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_B T} dp_x \right] \left[\int_{-\infty}^{\infty} e^{-p_y^2/2mk_B T} dp_y \right] \left[\int_{-\infty}^{\infty} e^{-p_z^2/2mk_B T} dp_z \right]$$

Gaussian Integral Again

$$q = \frac{V}{h^3} \left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_B T} dp_x \right]^3$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha},$

$$q = \frac{V}{h^3} \left[\sqrt{2\pi mk_B T} \right]^3 = V \left(\frac{2\pi mk_B T}{h^2} \right)^{3/2}$$

Q of Perfect Gas

$$q = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}$$

For indistinguishable gas particles,

$$Q = \frac{1}{N!} q^N = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Helmholtz Energy

$$Q = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Now, we can calculate the Helmholtz energy:

$$A = -k_B T \ln Q = -k_B T \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} \right\}$$

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Internal Energy

$$Q = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Also the internal energy:

$$U = -\frac{\partial(\ln Q)}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} \right\}$$

$$U = \frac{\partial}{\partial \beta} \ln \beta^{3N/2} + \frac{\partial}{\partial \beta} (\text{terms independent to } \beta) = \frac{3}{2} N k_B T$$

Pressure

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Using $p = -(\partial A / \partial V)_{T,n}$,

$$p = Nk_B T \frac{\partial(\ln V)}{\partial V} + \frac{\partial}{\partial V} (\text{terms independent to } V)$$

$$p = \frac{Nk_B T}{V}$$

One can obtain **entropy** and **chemical potential** similarly

Real Gas

Particles are no more independent to each other

$$Q \neq q^N / N!$$

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

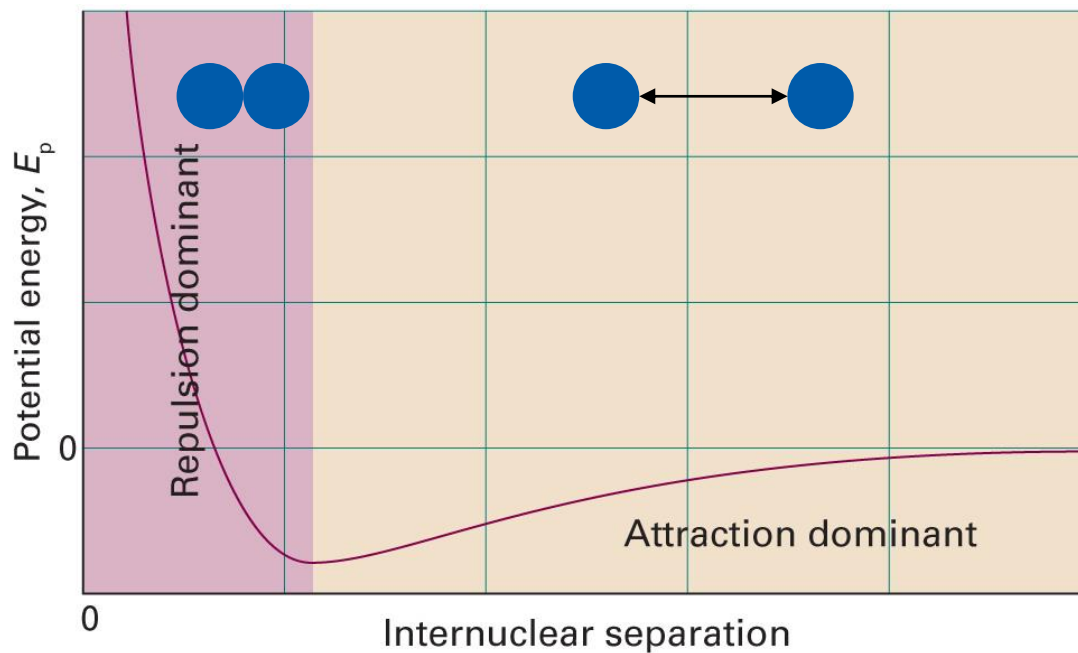
where

$$E_{\text{tot}} = \sum_i \frac{p_i^2}{2m} + U(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$$

Not in Atkins

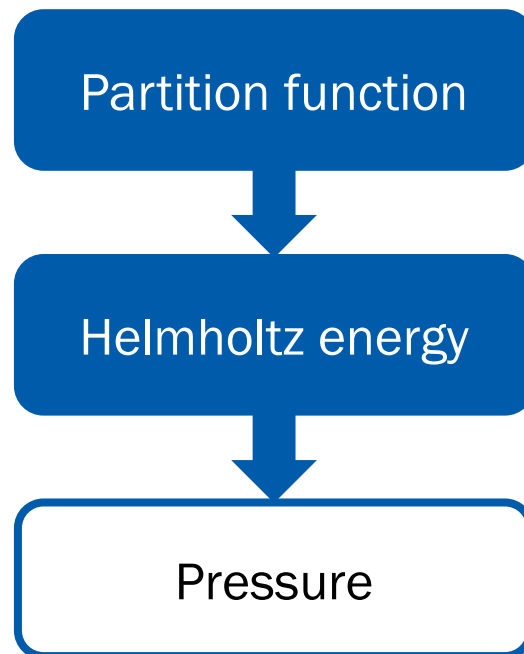
Potential Energy

Pairwise interactions (attraction + repulsion)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Pressure of Real Gas



$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each **quadratic contribution**
to the internal energy is $(1/2)k_B T$.

(Quadratic contribution: e.g., $E \propto p_x^2$ or $E \propto x^2$)

Can we prove this theorem?

Setting up the Stage

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_{\text{B}}T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Assume that

$$E_{\text{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$
$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

Summation to Product

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-\left(\sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2\right)/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

$$= \frac{1}{N! h^{3N}} \int \cdots \int \prod_{i=1}^{f_r} e^{-A_i r_i^2/k_B T} \prod_{j=1}^{f_p} e^{-B_j p_j^2/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Thus,

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2/k_B T} dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_B T} dp_j$$

Gaussian Integrals

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2 / k_B T} dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2 / k_B T} dp_j$$

Using

$$\int_{-\infty}^{\infty} e^{-A_i r_i^2 / k_B T} dr_i = \sqrt{\frac{\pi k_B T}{A_i}}, \quad \int_{-\infty}^{\infty} e^{-B_j p_j^2 / k_B T} dp_j = \sqrt{\frac{\pi k_B T}{B_j}},$$

we obtain

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_B T}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_B T}{B_j} \right)^{1/2}$$

In Terms of β

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_B T}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_B T}{B_j} \right)^{1/2}$$

We can write the partition function as follows:

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta} \right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j} \right)^{1/2}$$

Internal Energy

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta}\right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j}\right)^{1/2}$$

The internal energy can be calculated from

$$U = - \left(\frac{\partial \ln Q}{\partial \beta} \right)$$

Now,

$$\ln Q = -\frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

Internal Energy

$$\ln Q = -\frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

$$U = -\left(\frac{\partial \ln Q}{\partial \beta}\right) = \frac{f_r + f_p}{2} \frac{\partial \ln \beta}{\partial \beta} = \frac{f_r + f_p}{2} \frac{1}{\beta} = \frac{f_r + f_p}{2} k_B T$$

Therefore,

$$U = (f_r + f_p) \frac{k_B T}{2}$$

“The average value of each **quadratic contribution**
to the internal energy is $(1/2)k_B T$.”

Summary

